

ZZ ILab Final Presentation: Factor Momentum

24.02.2026

*Giulio Iepure, Caterina Piacentini,
Farkas Tallos, Tomas Samaj*



--- Agenda

1

Introduction – Factor Momentum

2

Part 1 (Factor level) – Performance & Robustness Checks

3

Part 2 (Single-stock level) – Replication, Net Positions, Frictions

4

Strategy Conclusion, Portfolio Implications & Implementation Constraints

5

R Shiny App Usage Demonstration

6

Ilab Conclusions: Lessons Learned & Empirical Challenges

Literature Review & Methodology

Arnott, Kalesnik & Linnainmaa (2023):

- Extends momentum research to factor portfolios – showing that factor returns themselves exhibit momentum.
- Finds that factor momentum subsumes industry momentum.
- Uses principal component analysis to identify systematic sources of momentum.

Ehsani & Linnainmaa (2022):

- Shows that momentum in stock returns stems from momentum in factor returns
- Factors show strong autocorrelation: winners stay winners, losers stay losers.
- Momentum reflects timing of factor exposures, not a separate risk factor.

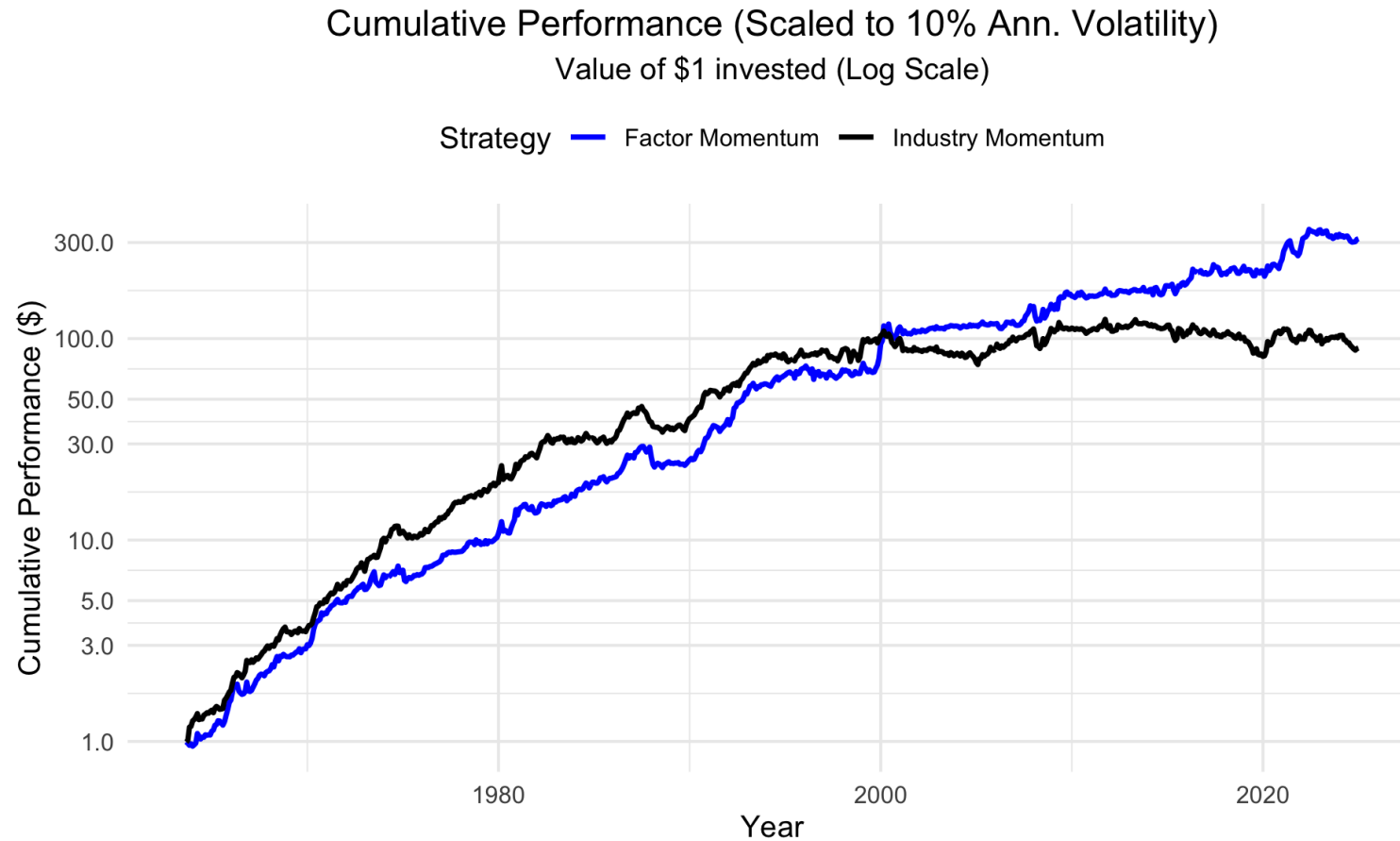
Data Description:

- **Fama-French 17 Industry Portfolios:** Monthly excess returns (July 1963 - Dec 2024).
- **JKP Factors:** Cross-sectional factor returns (value, profitability, investment, quality, risk).
- **Thematic Factors:** Growth, leverage, volatility, and other economic themes.
- **Final Sample:** July 1963 - December 2024 (U.S. market).

Replication Methodology:

- Replicate Arnott et al. (2023) Appendix figure comparing **Factor vs. Industry Momentum**.
1. For each month:
 - Rank all assets by previous month's return.
 - Go **long** the top half (winners), **short** the bottom half (losers).
 2. Apply to Fama-French 17 industries and selected JKP factors.
 3. Equal-weight portfolios; rebalance monthly.
 4. Compute 1-month long-short return.

Replication Results



I

Volatility Scaling

To match the paper's methodology, the strategies are scaled to a **target volatility of 10% per year**.

II

Before 2000

Industry momentum consistently outperforms factor momentum, showing stronger cumulative returns.

III

After 2000

Industry momentum stops growing, while **factor momentum accelerates and becomes the clear winner**.

IV

... result in a structural shift

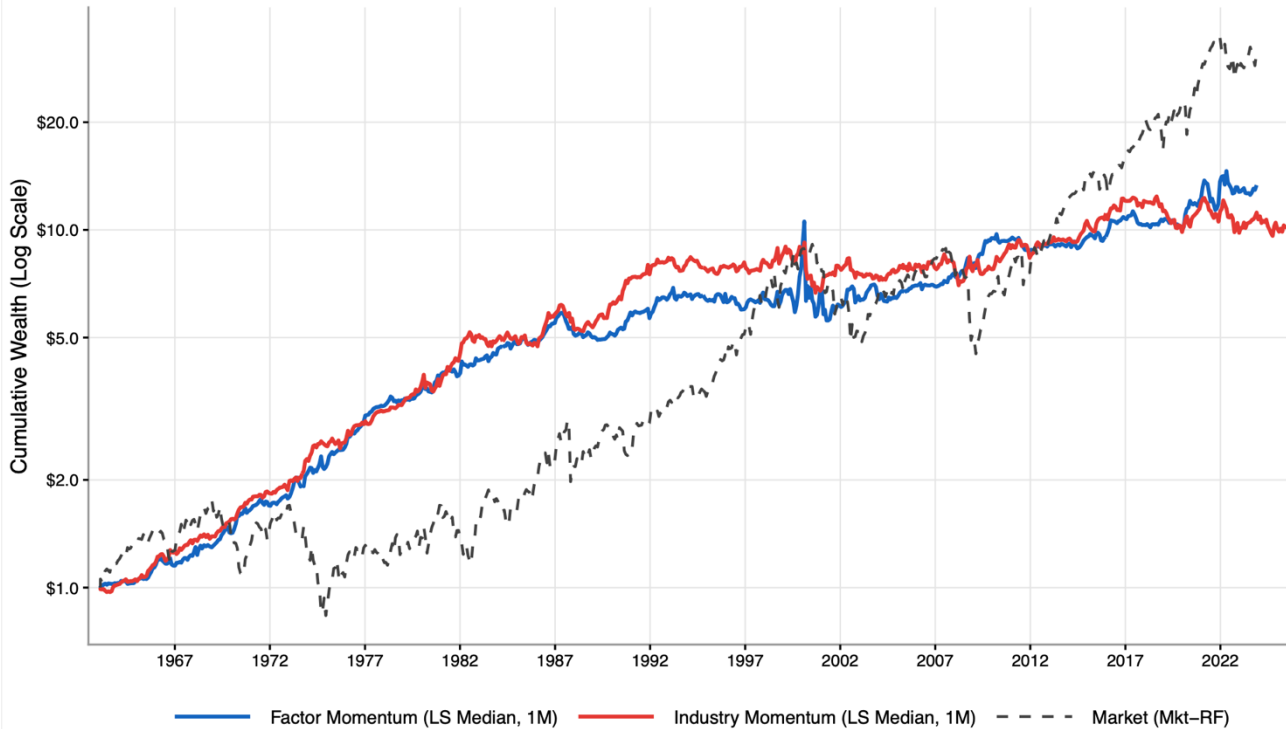
This trend is even more pronounced in the unscaled strategy.

Replication Results

We add a **1-day Implementation Lag** to make the strategy more realistic.

Factor Momentum vs. Industry Momentum vs. Market

Cumulative Wealth, \$1 Invested in 1963 | Log Scale | LS Median, 1–Month Lookback | 1–Day Implementation Lag



We compare factor momentum and industry momentum against the **Fama-French market factor (Mkt-RF)** to benchmark their long-run cumulative performance and relative behavior across market regimes.

Industry momentum broadly tracks the direction of the other series but delivers **weaker long-run cumulative growth**, indicating lower upside over the full sample.

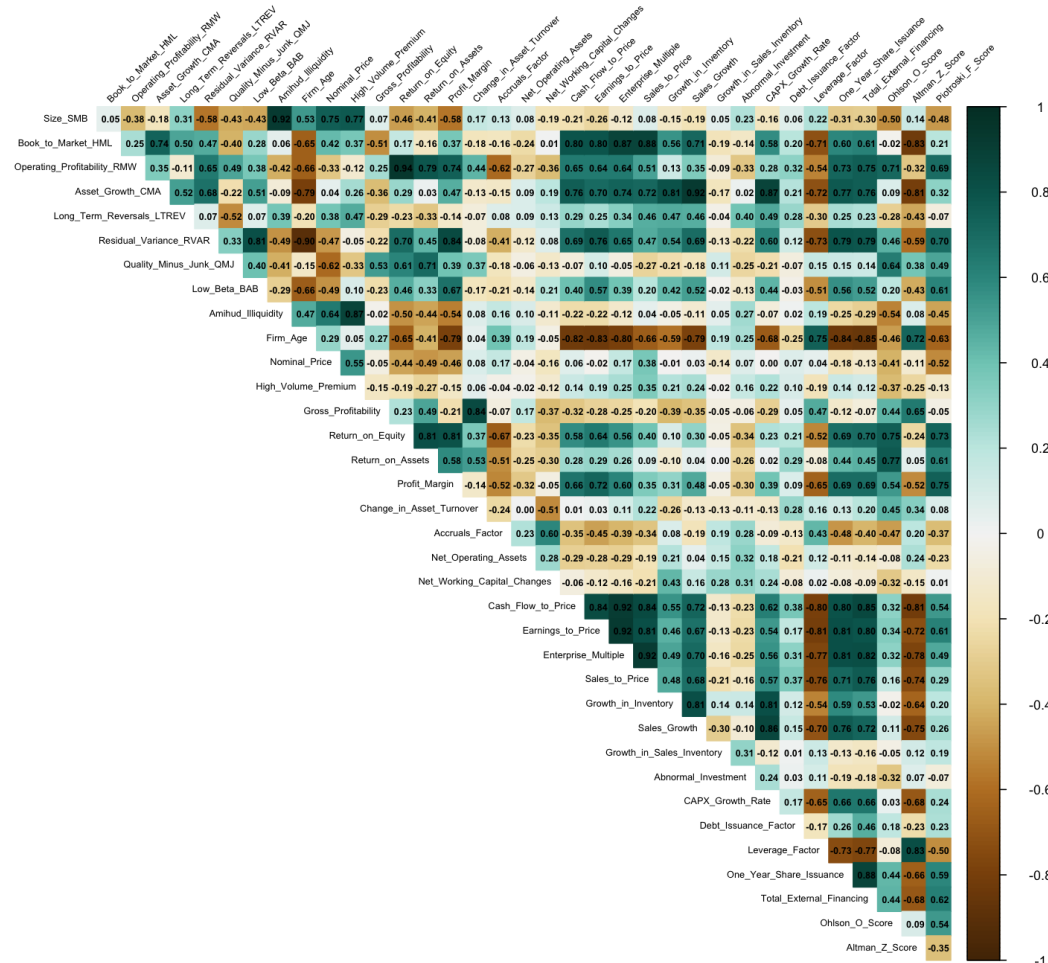
The market factor (Mkt-RF) delivers the **strongest long-run cumulative growth** in the sample, but with a more pronounced cyclical path and deeper drawdowns.

Factor momentum outperforms industry momentum and remains **closer to the market** factor over the long run, with a smoother trajectory.

Overall, factor momentum stands out as the strongest alternative to market, while industry momentum behaves more defensively but with limited upside.

Factor Correlation Heatmap

Upper Triangular Correlation Heatmap of Selected Factors



Correlation across JKP Factors

We observe clusters of **highly correlated factors (dark green)**, such as profitability and investment factors, as well as groups **with low or negative correlation (brown)**.

Uncorrelated Factors

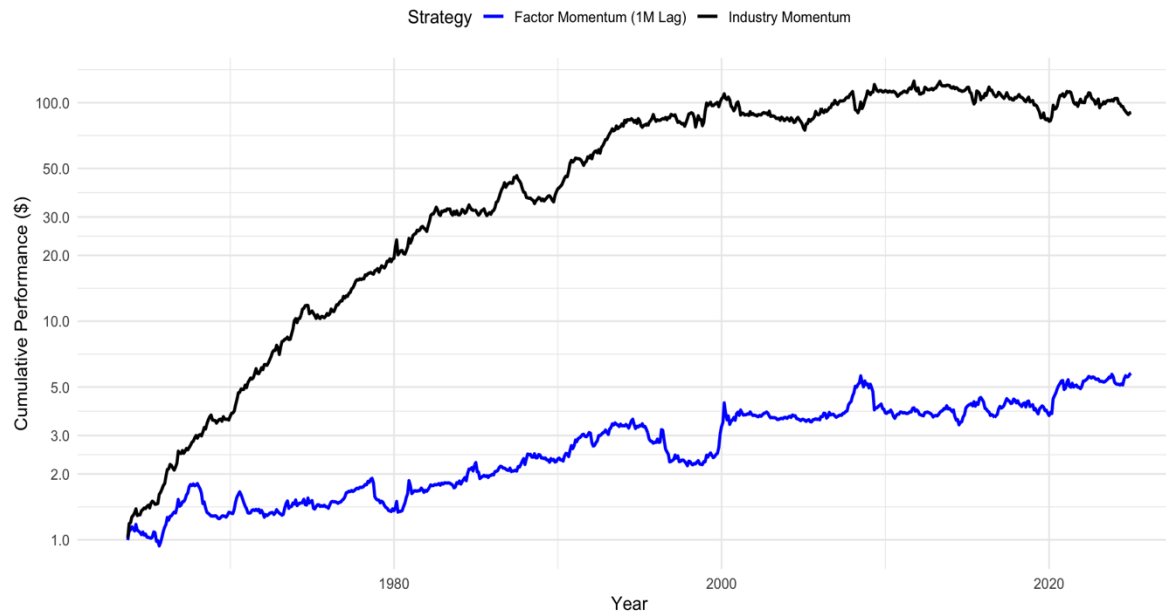
Size (SMB) and momentum-related factors remain largely orthogonal, reflecting their distinct economic drivers.

Effects on Momentum

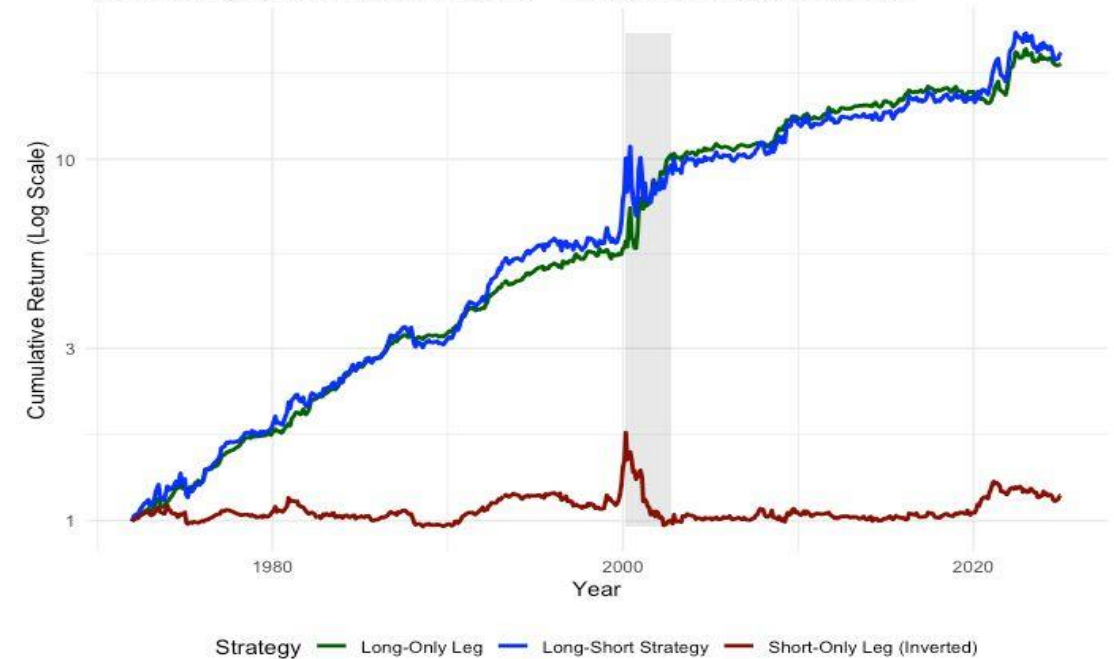
Fundamental factors move together, reinforcing systematic momentum.

Robustness Checks

Cumulative Performance (Scaled to 10% Ann. Vol): Industry (T) vs Factor (T+1)
Value of \$1 invested (Log Scale)



Factor Momentum: Long vs. Short Leg Performance (Arnott et al. 2023 Replication)
1M/1M Strategy, Median Split. Shaded area = 2000-2002 Dot-com bubble burst.



Lagging



Lagging the signal by 1 month kills the performance.

Long/Short Performance



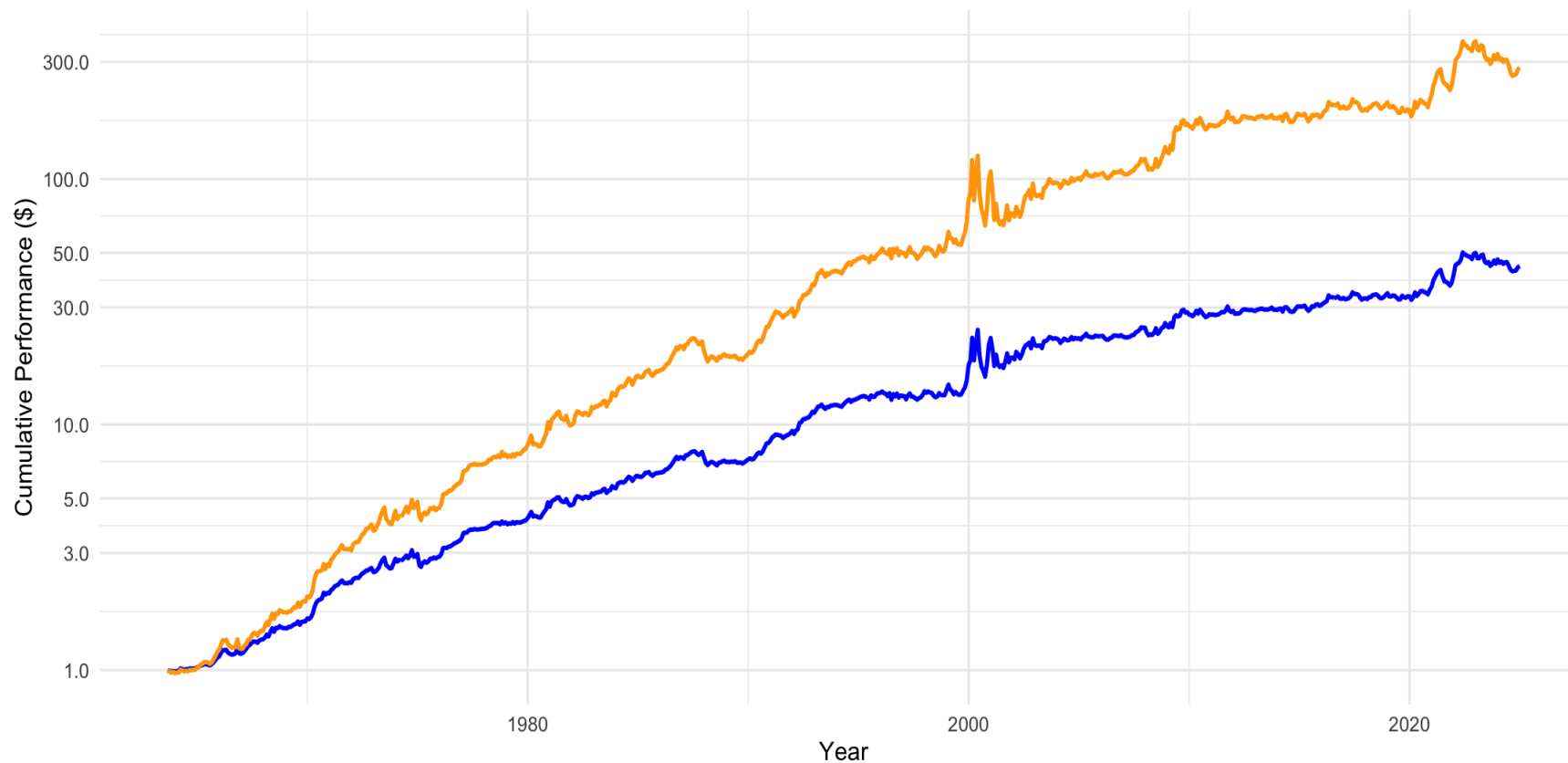
Most of the performance of the strategy comes from the long leg.

Change of Momentum Definition

Factor Momentum Comparison (Raw Returns)

50/50 (Median) vs. 25/25 (Quartile) Strategy | Log Scale

Strategy — Factor Momentum — Factor Momentum (Quartile)



We changed the definition of momentum to **25%/25%**: the orange line is **long top 25% and short bottom 25%**.



Tail-sorting strengthens the signal: focusing on **extreme winners/losers** widens the spread and boosts momentum.



Best absolute performance is long-short 25%: the top-25% long-short portfolio delivers the highest standalone returns.

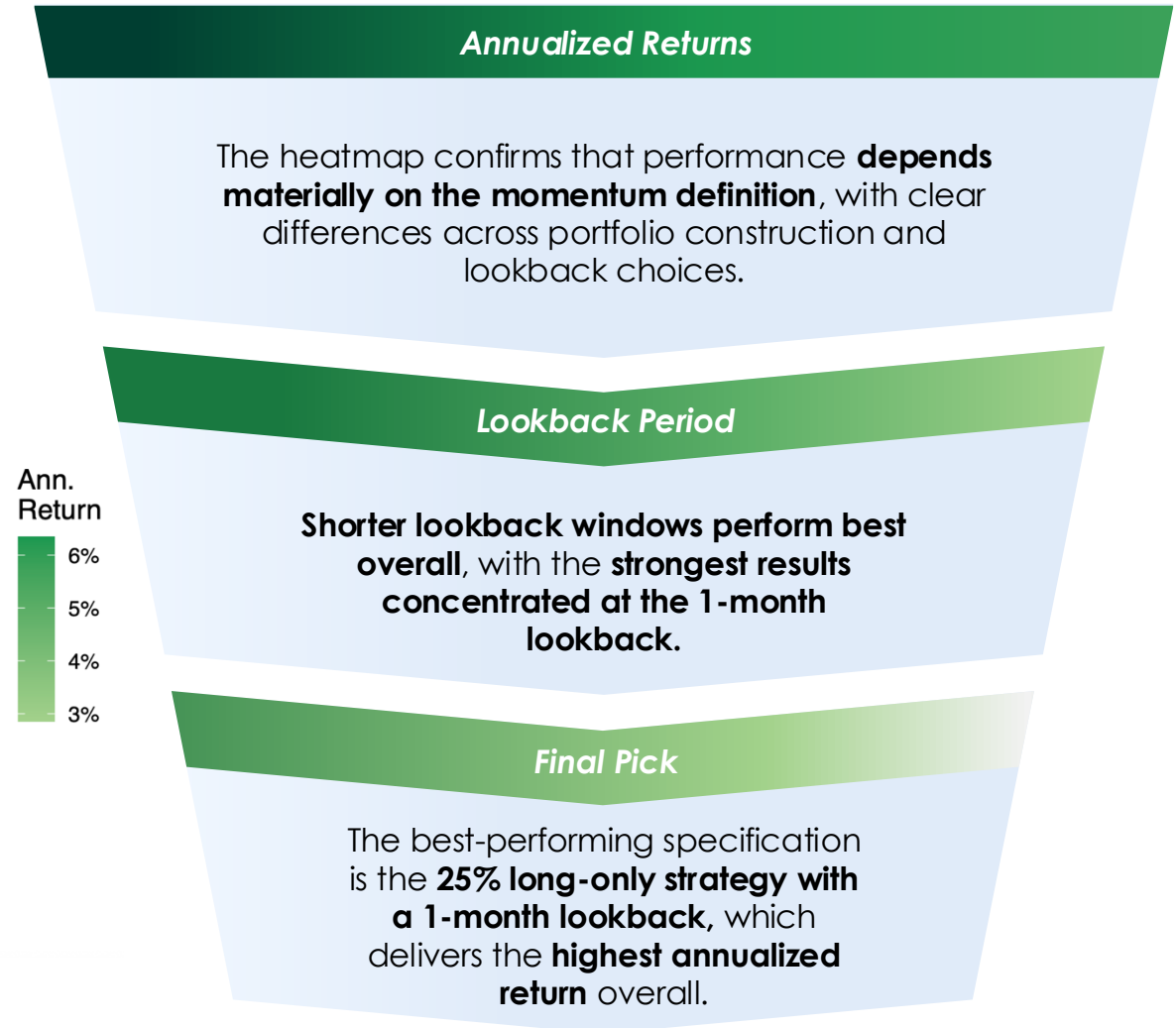
Change of Momentum Definition

Factor Momentum: Annualized Return Across Parameter Grid

JKP Daily Factors | 1963– | 1–Day Implementation Lag | 21–Day Holding Period

Strategy Type	Lookback Window				
	1M	2M	3M	6M	12M
LO_25	6.4%	5.8%	5.2%	5.6%	5.8%
LO_33	6.1%	5.6%	5.1%	5.3%	5.6%
LO_Median	5.4%	4.9%	4.7%	4.8%	4.9%
LS_25	6.3%	5.3%	4.1%	4.7%	4.9%
LS_33	5.8%	4.6%	3.8%	4.2%	4.3%
LS_Median	4.4%	3.4%	2.9%	3.0%	3.1%

LS = Long–Short | LO = Long–Only | Median / 33% / 25% split



Change of Momentum Definition



The original **LS median** specification is the worst-performing strategy in **every** lookback window.

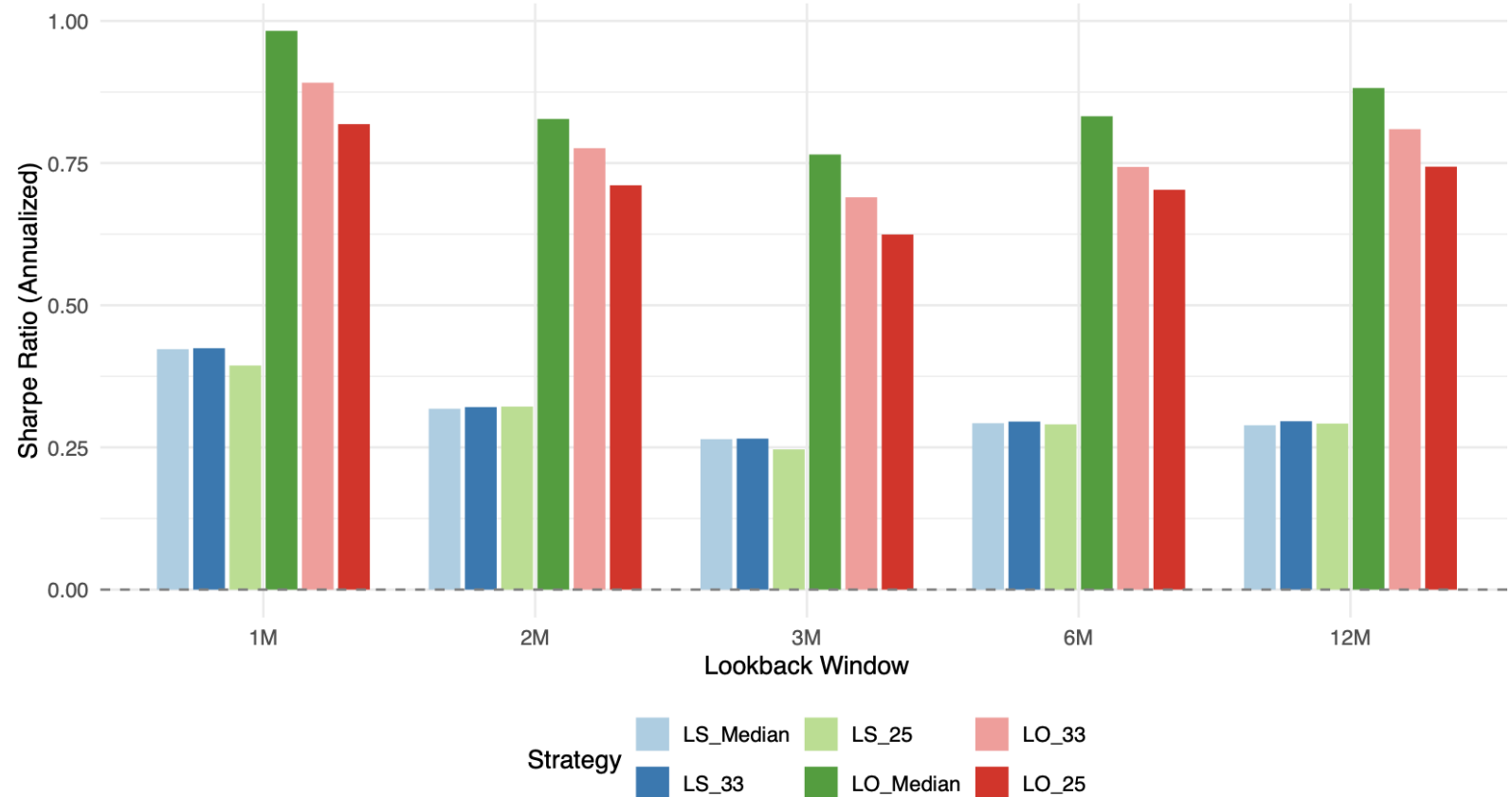


Across all lookback windows, **long-only specifications consistently outperform the corresponding long-short alternatives**, indicating that the short leg weakens overall performance.



Even though the **long-only top 25%** delivers the highest returns, it also comes with **higher volatility**, which **reduces its risk-adjusted performance** relative to others long-only alternatives.

Sharpe Ratio by Strategy Type and Lookback Window
Factor Momentum | JKP Daily | 1963– | 1–Day Lag

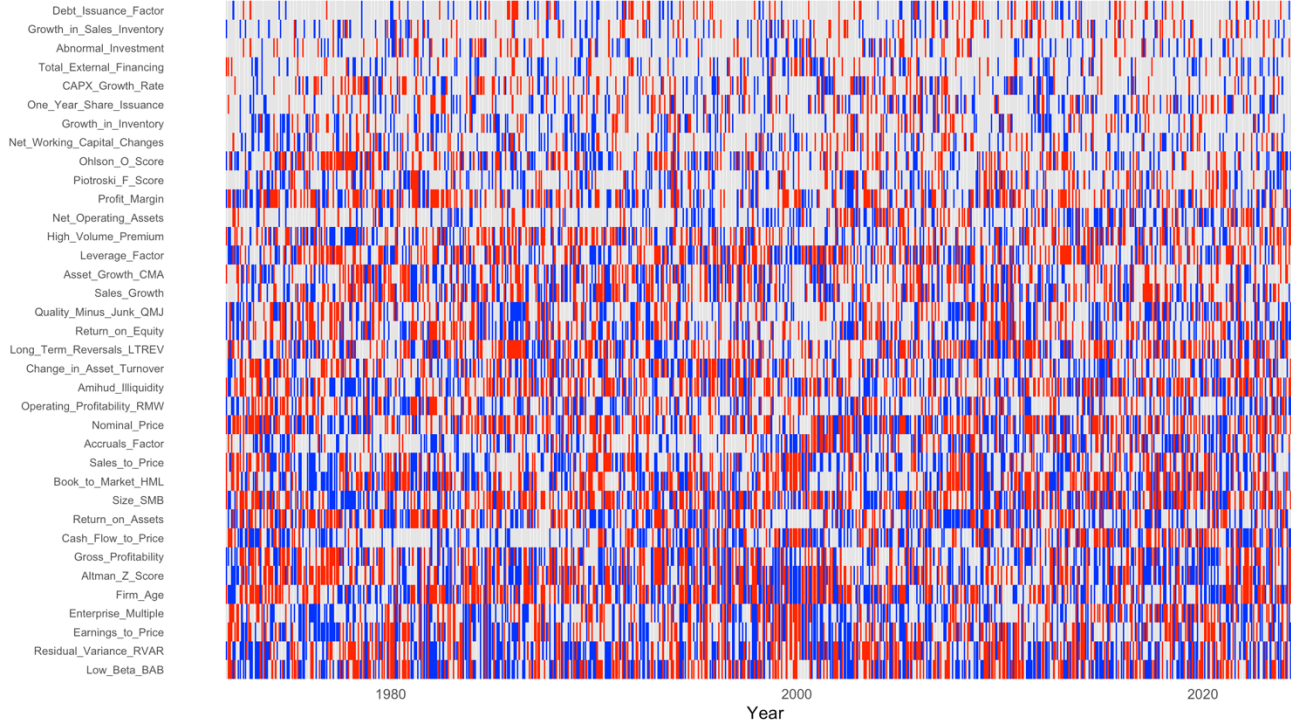


Is it Possible to Reduce the Number of Factors?

Factor Positions Over Time (Quartile Split)

Blue = Top 25% (Long), Red = Bottom 25% (Short), Grey = Middle 50% (Unselected)

Position ■ Long ■ Neutral ■ Short



The Heatmap:

Factors move in and out of the top and bottom quartiles frequently, so there is no stable subset of 'always-long' or 'always-short' factors.

Long Factors:

Top 5 Long factors are **Low_Beta_BAB** (37.0%), **Residual_Variance_RVAR** (36.7%), **Earnings_to_Price** (32.4%), **Enterprise_Multiple** (32.4%), and **Firm_Age** (32.0%).

Short Factors:

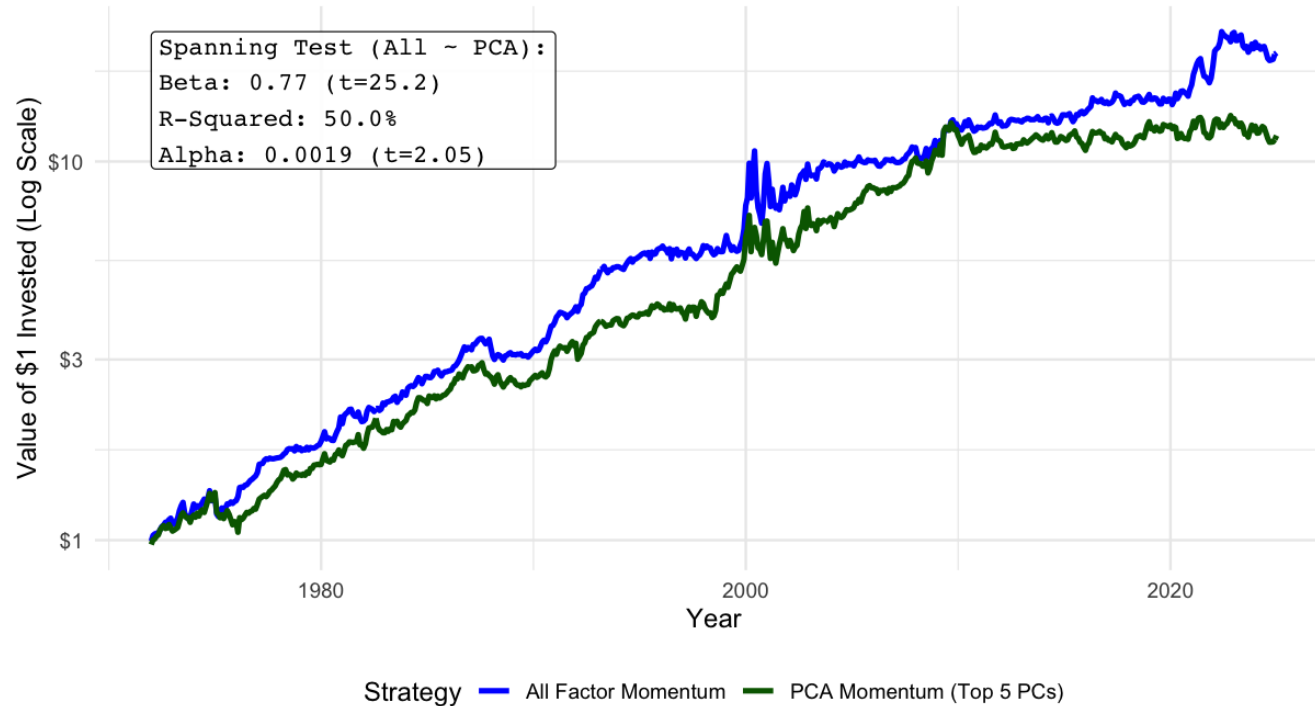
Top 5 Short factors are **Nominal_Price** (38.9%), **Size_SMB** (38.2%), **Firm_Age** (37.5%), **Low_Beta_BAB** (37.1%), and **Altman_Z_Score** (35.9%).

The factor universe cannot be reduced in a stable way. Rankings are highly time-varying, and even leading candidates frequently switch sign (e.g., Low_Beta_BAB).

Systematic Validation: Top 5 Principal Components Drive the Strategy

Systematic Validation: PCA vs. All Factors

Top 5 Principal Components capture the majority of the signal

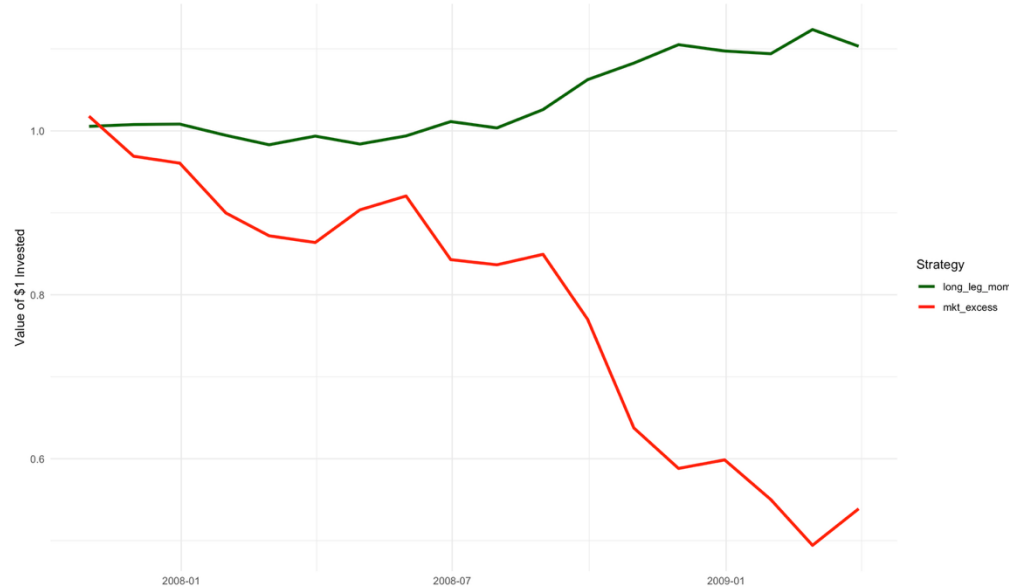


The PCA decomposition shows that the **top 5 principal components** explain more than **80% of total variance** in factor returns. This means the factor universe is **largely driven by a small set of common, time-varying systematic forces**, while the **role of individual factors changes over time**. As these dominant exposures rotate, the PCA results do not support a stable pre-selection of “best” factors.

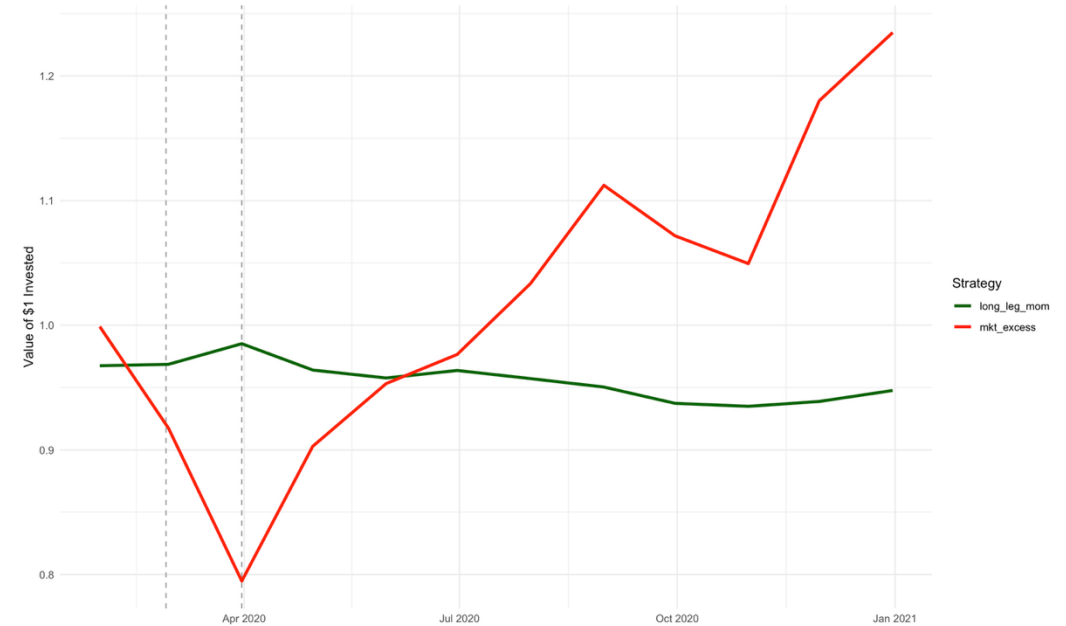
We proceed with the rest of the analysis using the full 35-factor universe.

Performance During Crisis (Long-Only)

GFC Crisis Trajectory: Long-Leg Factor Mom vs Market
Cumulative Performance (Oct 2007 - Mar 2009)



COVID 2020 Trajectory: Long-Leg Factor Mom vs Market



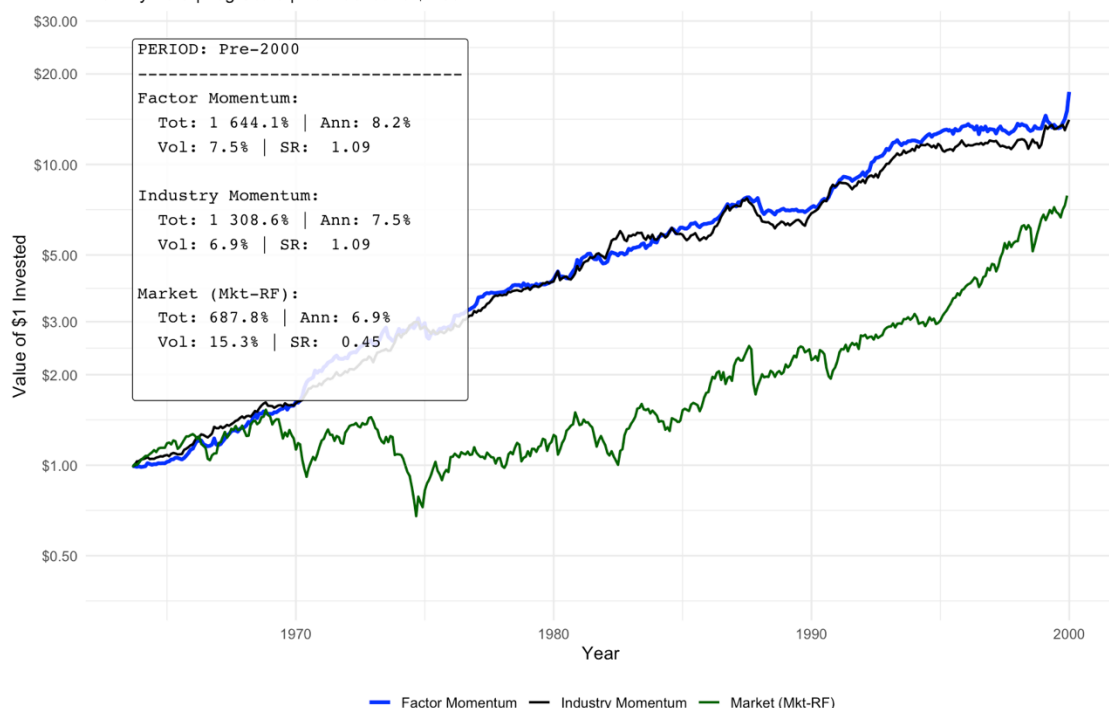
Period	Long Leg Return	Market Return	Correlation	Long Leg Volatility	Market Volatility
Covid	0.0182	-0.204	-1	0.039	0.128
GFC	0.103	-0.461	- 0.562	0.0541	0.221

Across crisis episodes, the strategy provides a consistent diversifying hedge, helping stabilize portfolio outcomes in drawdowns.

Structural Decay in the Performance Post-2000

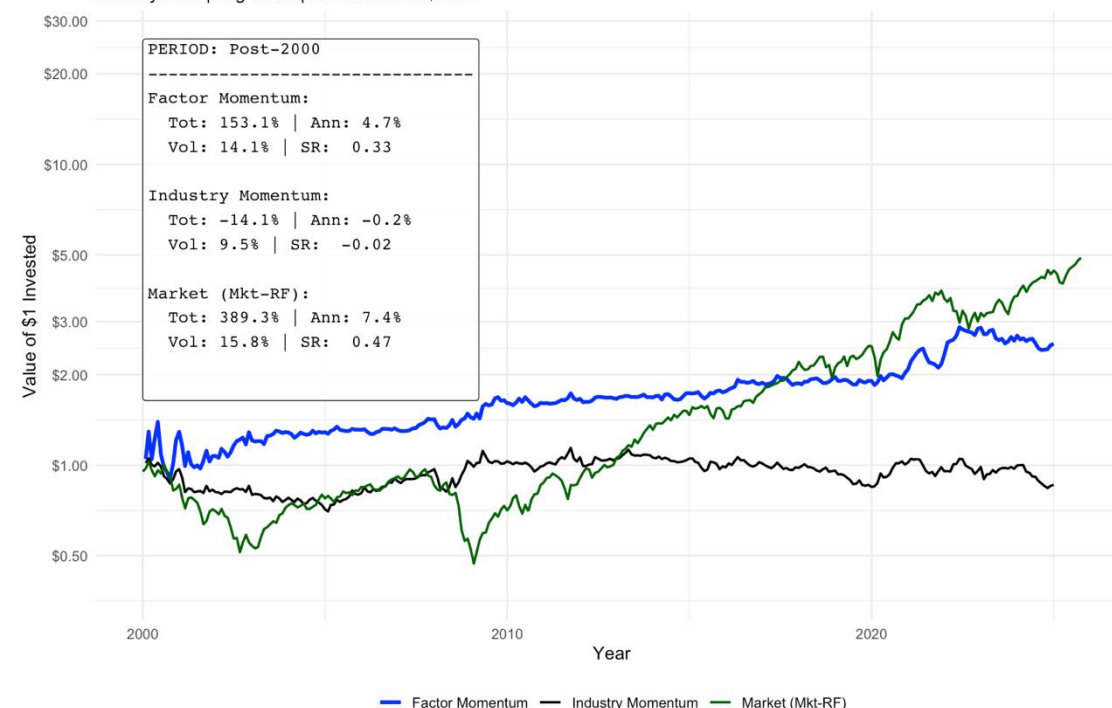
Strategy Comparison: Pre-2000

Monthly Data | Log Scale | Re-indexed to \$1.00



Strategy Comparison: Post-2000

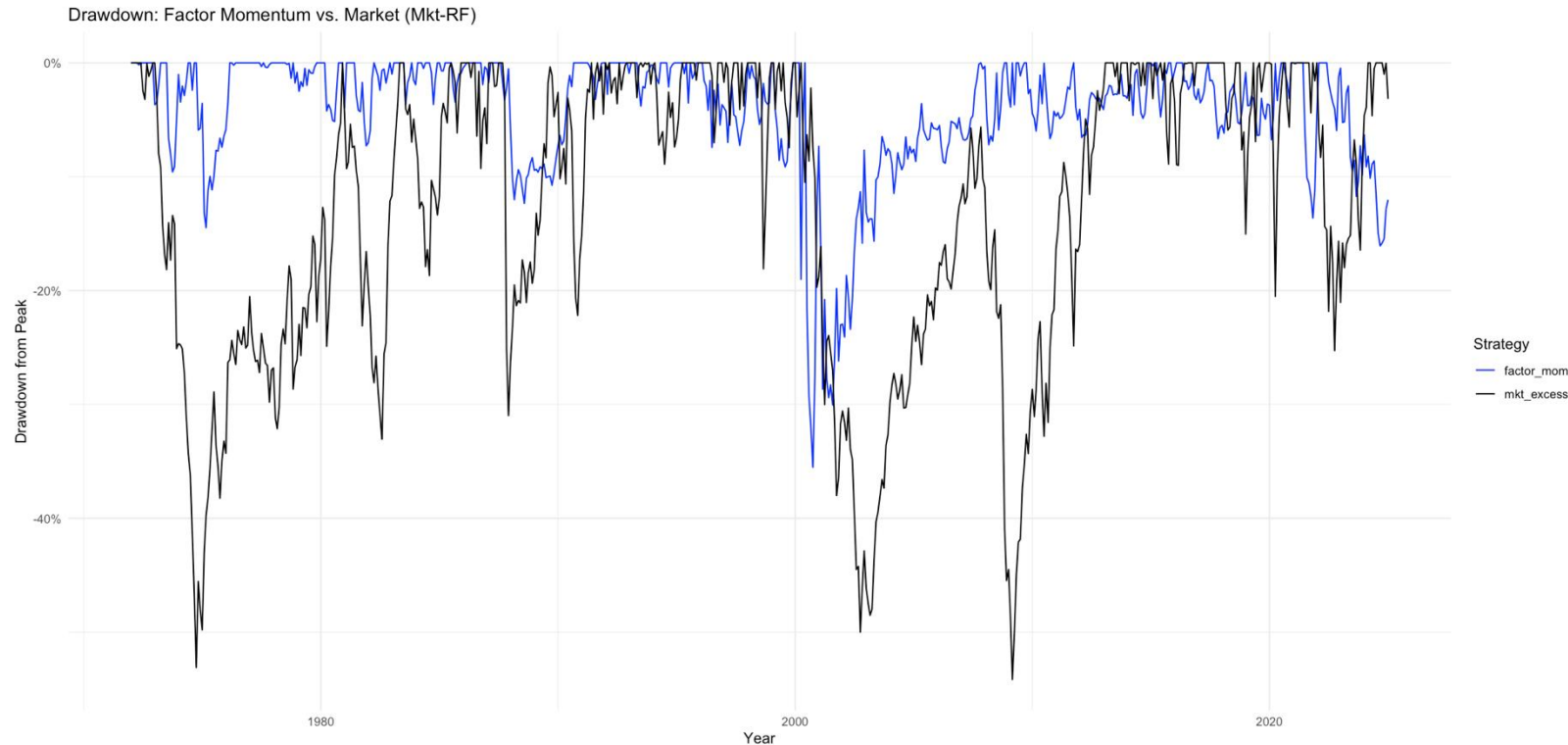
Monthly Data | Log Scale | Re-indexed to \$1.00



Why Alpha Has Decayed Post-2000: Structural alpha decay in the modern era (from SR=1.09 to 0.33) is probably driven by increased market efficiency and the crowding of factor strategies following widespread academic publication.

To restore investability, in Phase 2 we pivot to 25% long-only implementation that leverages the strategy's proven negative correlation during market stress as a reliable tail-risk hedge

Structural Decay in the Performance Post-2000



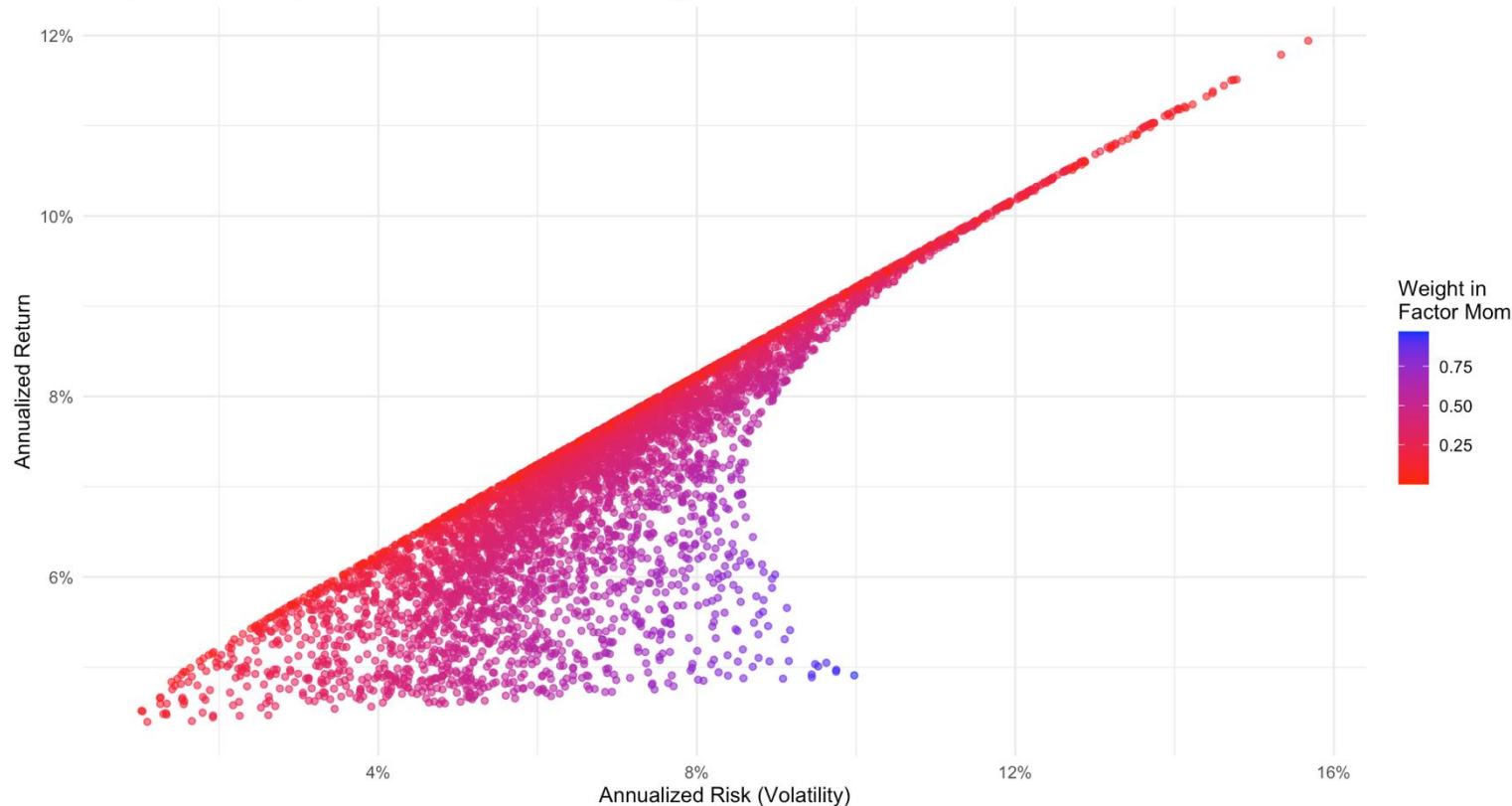
Why Alpha Has Decayed Post-2000: Structural alpha decay in the modern era (from $SR=1.09$ to 0.33) is probably driven by increased market efficiency and the crowding of factor strategies following widespread academic publication.

To restore investability, in Phase 2 we pivot to 25% long-only implementation that leverages the strategy's proven negative correlation during market stress as a reliable tail-risk hedge

Portfolio Impact: Factor Momentum Acts more as a Volatility Dampener

Efficient Frontier Analysis: Adding Factor Momentum

Blue points indicate high allocation to Factor Momentum strategy.



The efficient frontier reveals that **portfolios with high allocations to Factor Momentum** (blue points) cluster in the **lower-volatility region**, sacrificing the raw upside of a pure equity portfolio (red points) to achieve a **significantly more stable risk profile**.

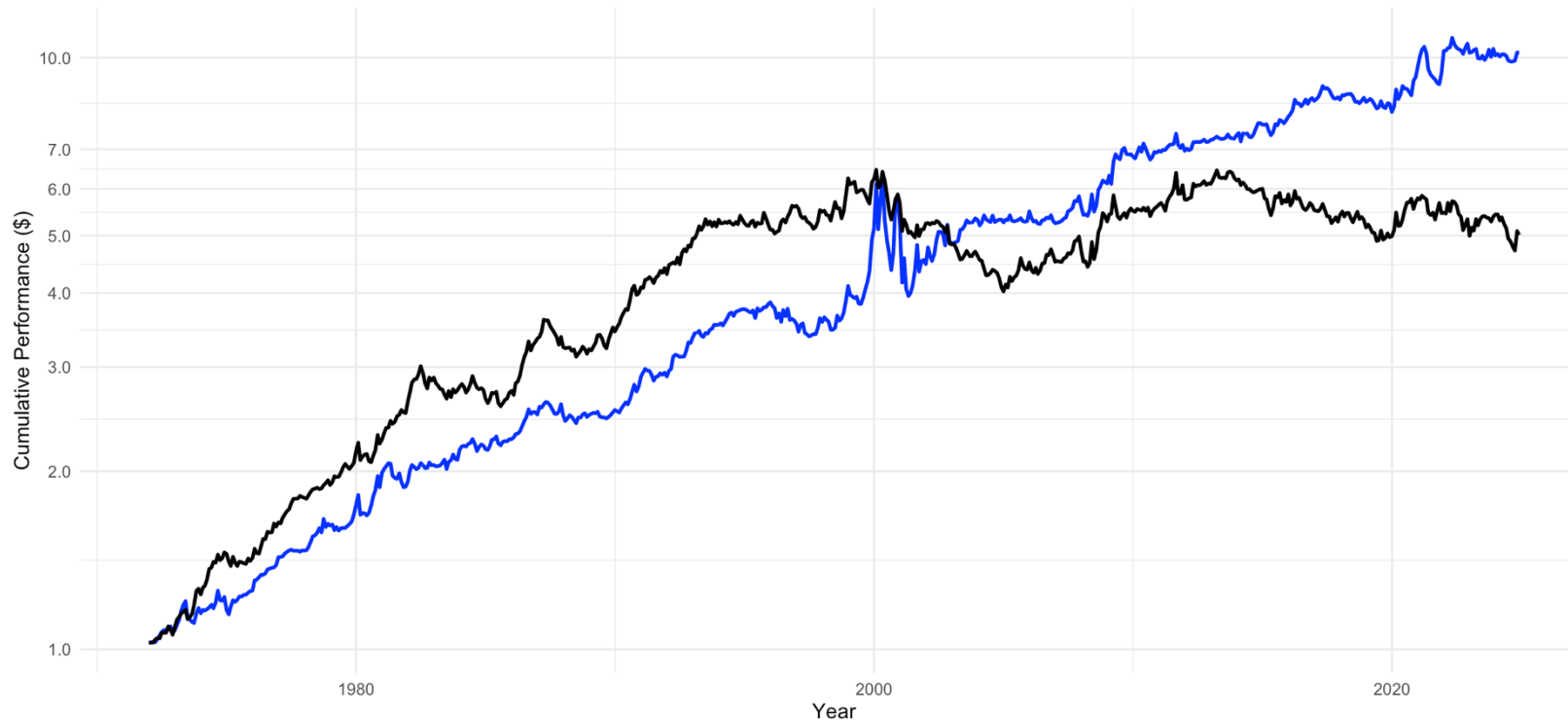
To restore investability, in Phase 2 we pivot to 25% long-only implementation that leverages the strategy's proven negative correlation during market stress as a reliable tail-risk hedge

Switching to Daily Data – Closer to Real-life implementation

Cumulative Performance: Factor vs. Industry Momentum (Both Lagged)

Daily Signal with 1-Day Implementation Lag applied to both strategies

Strategy — Factor Momentum (1-Day Lag) — Industry Momentum (1-Day Lag)



We switched to **daily data** to mimic real-life implementation of the Investment Strategy.



Replication - daily data: instead of investing at EoM close prices, invest at **1st day of the following month closing prices**.

Going To Part Two

1) Rejection of Pure Replication (Long/Short):

- We advise against deploying the fully replicated Long/Short Arnott et al. (2023) strategy as a standalone alpha generating strategy.
- **Reasoning:** The structural alpha decay post-2000 (Sharpe Ratio declining from 1.09 to 0.33) implies that the strategy, in its raw form, likely cannot overcome the transaction costs associated with high-turnover shorting.

2) Strategic Pivot: Long-Only Implementation:

- We recommend isolating the **25% Long-Only leg** for implementation.
- **Rationale:** The results show that the 25% Long-Only definition delivers the strongest performance, and the replication further confirms that most of the cumulative return is driven by the long leg, while the short leg mainly adds volatility without contributing significant returns.
- **Benefit:** This choice preserves the upside driver in the framework while significantly reducing implementation costs and improving practical scalability.

3) Portfolio Role - Crisis Diversifier:

- **Sizing:** The strategy should be sized as a tail-risk hedge rather than a core equity replacement.
- It demonstrates unique resilience during market stress, delivering returns during the GFC (+2.8%) and the COVID-19 crash (+7.2%).
- **Correlation:** -0.034 to the 60/40 portfolio → it provides genuine diversification benefits.

4) Execution Constraints:

- **Universe:** The attempt to select a "unique" subset of factors resulted in underperformance, suggesting that broad diversification is required to capture the momentum premium effectively.
- **Timing:** Our robustness checks show that a 1-month lag completely erodes the outperformance, while the 1-day lag harms it significantly, necessitating a strict rebalancing schedule.

Factor Mapping and Replication from Single Stocks

The JKP (2025) Framework

- **Phase 1:** Mapped 36 characteristics to the *Factor Momentum* literature using the **Jensen, Kelly, and Pedersen (JKP)** framework.
- **The Breakthrough:** After initial challenges finding suitable code and struggling with replication of other factors than the FF from Tidy Finance, we recently identified the **official JKP Python replication repository**, released in **December 2025**.

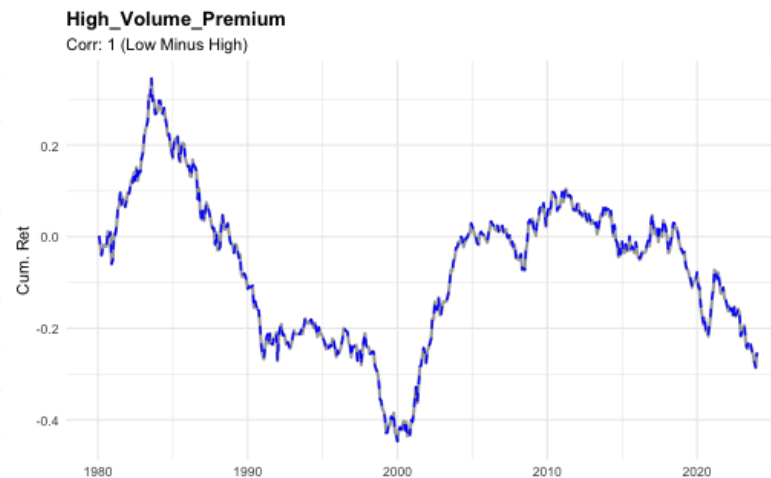
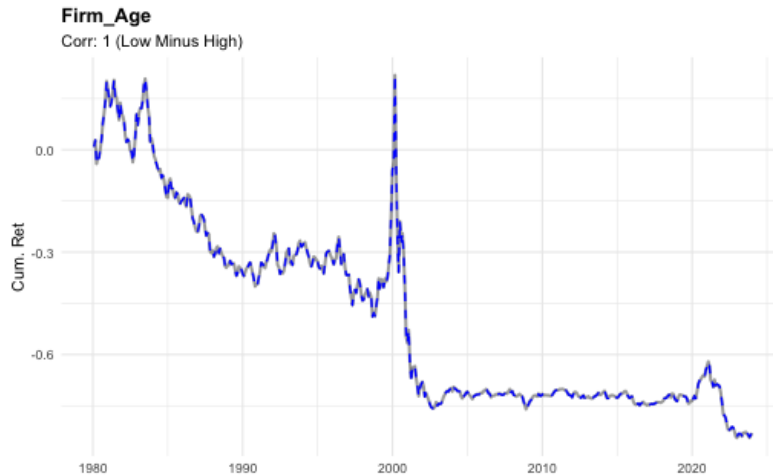
Implementation

- **Infrastructure:** Leveraged the **WU Cluster** to handle the high computational load (matching JKP's requirement of 450GB RAM / 128 CPU cores).
- **Status:** Successfully replicated all factors with near-exact benchmark matches through **Year-End 2024**.
- **Implemented Update:** Adapted to a **new CRSP data structure** via a JKP update released in January 2026.

Factor Name	Variable Key	Factor Name	Variable Key
Size_SMB	market_equity	Net_Working_Capital_Changes	cowc_gr1a
Book_to_Market_HML	be_me	Cash_Flow_to_Price	ocf_me
Operating_Profitability_RMW	ope_be	Earnings_to_Price	ni_me
Asset_Growth_CMA	at_gr1	Enterprise_Multiple	ebitda_mev
Long_Term_Reversals_LTREV	ret_60_12	Sales_to_Price	sale_me
Residual_Variance_RVAR	ivol_ff3_21d	Growth_in_Inventory	inv_gr1
Quality_Minus_Junk_QMJ	qmj	Sales_Growth	sale_gr1
Low_Beta_BAB	betabab_1260d	Growth_in_Sales_Inventory	dsale_dinv
Amihud_Illiquidity	ami_126d	Abnormal_Investment	capex_abn
Firm_Age	age	CAPX_Growth_Rate	capx_gr1
Nominal_Price	prc	Debt_Issuance_Factor	dbnetis_at
High_Volume_Premium	dolvol_126d	Leverage_Factor	at_be
Gross_Profitability	gp_at	One_Year_Share_Issuance	chcsho_12m
Return_on_Equity	ni_be	Total_External_Financing	netis_at
Return_on_Assets	niq_at	Ohlson_O_Score	o_score
Profit_Margin	ebit_sale	Altman_Z_Score	z_score
Change_in_Asset_Turnover	at_turnover	Piotroski_F_Score	f_score
Accruals_Factor	oaccruals_at	Net_Operating_Assets	noa_at

Factor Replication (Example)

Audit Grids - Page 1



Factor Momentum Strategy Replication

1) Replication of Factors

- We replicate all **36 factors** using single stock data from WRDS (CRSP, Compustat).
- Weights of single stocks (long/short positions) are saved within the factor construction process.

2) Factor Momentum Strategy

- The Arnott et al. (2023) Factor Momentum strategy is constructed from the replicated factors, similarly to the 1st part.
- We proceed with the **Long-only 25%** Factor Momentum Strategy, as it yielded the best results in the 1st part.

3) Net Positions

- Net positions for the Factor Momentum strategy are calculated on the **single stock level** across factors.
- As a certain stock can have a long positions in one factor, but a short position within another, it is crucial to net out and work with the resulting position before implementing frictions.

4) Frictions

- We implement the following frictions to mimic real-life implementation of the strategy:
 - 1) **Transaction Costs.**
 - 2) **Shorting Costs.**
 - 3) **Withholding tax on Dividends.**

Factor Momentum - Position Analysis

Number of stocks per month Plot



Transaction Costs

- Transaction costs are modelled in **basis points** as a decreasing function of **market capitalization**.
- We choose slope so that a representative large-cap has 5-10 bps one-way implementation shortfall, then let costs rise smoothly for smaller caps.

$$TC_i^{bps} = base^{bps} + slope \left(\frac{ref_{mcap}}{\max(MC_i, 1)} \right)^\alpha$$

Parameter	Value
$base^{bps}$	5 bps
$slope$	1 bps
α	0.25

- TC_i^{bps} : estimated one-way transaction cost (bps) for stock i .
 - $base^{bps}$: fixed min. cost level (bps) applied to very liquid names.
 - $slope$: calibrated sensitivity parameter (bps).
 - ref_{mcap} : median market cap. of the S&P 500.
 - α : curvature parameter that controls how quickly costs rise ($\alpha > 0$).
 - $\max(MC_i, 1)$: numerical guard to avoid division by zero.
- Max. TC is **capped at 100 bps** to avoid extreme values.

This allows us to approximate transaction costs using market capitalization for real world application.

Shorting Costs

- **D'Avolio (2002)** “The Market for Borrowing Stock” (*Journal of Financial Economics*).
 - On the short side, the main additional friction is the stock borrow fee. D'Avolio (2002) measures this as the implied loan fee - the difference between the Fed Funds rate and the rebate rate paid on the short-sale collateral.
 - **Key takeaway:** for large, liquid U.S. stocks, borrow costs are usually small: about 91% of loans have fees below 1% per year, and the value-weighted average is roughly **17 bps** per year for general collateral, about **24 bps** per year overall.
- **Adjustment:** we want to penalize small stocks with greater relative shorting costs, therefore we use a similar function as for TCs to calculate the final SCs as a function of market cap.

We incorporate borrow fees as an annualized charge on short notional (adjusted for market cap), as most liquid U.S. equities are cheap to borrow.

US-Austria Cross-Border Equity Tax Frictions

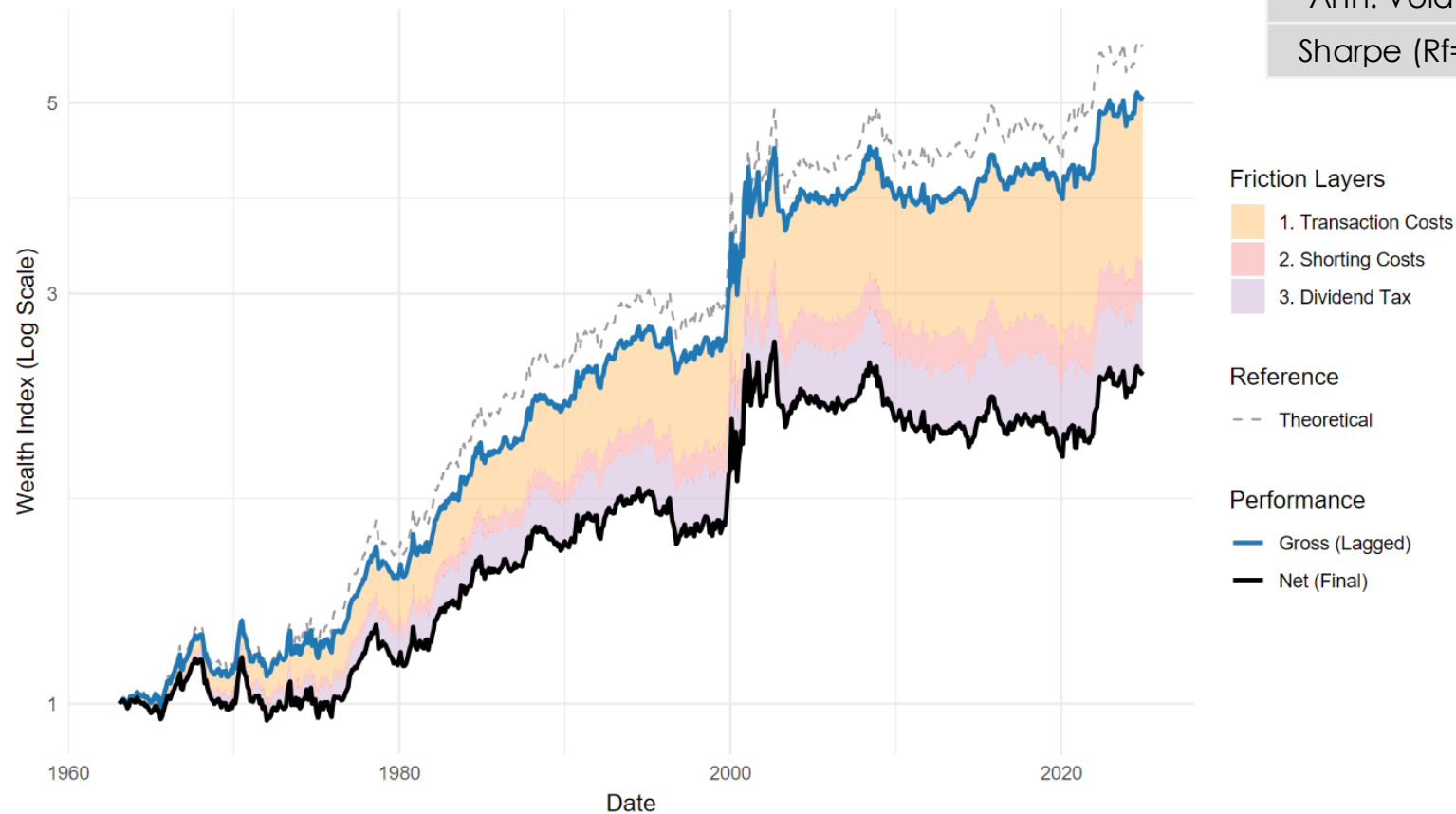
- **Bilateral Treaty Efficiency (Double Taxation Convention 1996):** Capital gains on US equities are taxed exclusively in Austria (0% US liability). Dividends benefit from a treaty-reduced **withholding tax of 15%** (Form W-8BEN required), which is fully creditable against the Austrian liability
- **The "Long" Leg Tax Drag:** All realized gains and dividends are subject to the Austrian **27.5%** special tax rate (Kapitalertragsteuer; Investment Income Tax). Post-2025 regulations mandate Tax-Base Netting; however, losses cannot be carried forward and expire annually.
- **The "Short" Leg Asymmetry:** Short positions face a "deadweight loss." Manufactured dividends paid to lenders are non-deductible for private investors due to the Austrian income-related expense deduction ban (§20 Abs 2 EStG), creating a significant "negative alpha" hurdle.
- **Strategy Implications:** Dividends on the long-leg side must be tax-adjusted to account for the tax differences of long/short positions.

We implement a parameter to asymmetrically tax dividends (27.5% rate) received on the long leg, while paying 100% of the manufactured dividends on the short leg.

Frictions

Strategy Performance: Decomposition of Frictions

Cumulative wealth lost to Transaction Costs, Borrow Fees, and Dividend Taxes



Strategy Metrics	Theoretical	Gross (Lagged)	Net (Final)
Ann. Return	0.0289	0.0266	0.0145
Ann. Volatility	0.0742	0.0724	0.0723
Sharpe (Rf=0%)	0.3901	0.3681	0.2004



- Transaction costs hurt the performance most (**high turnover**).
- Shorting costs seem to be manageable.

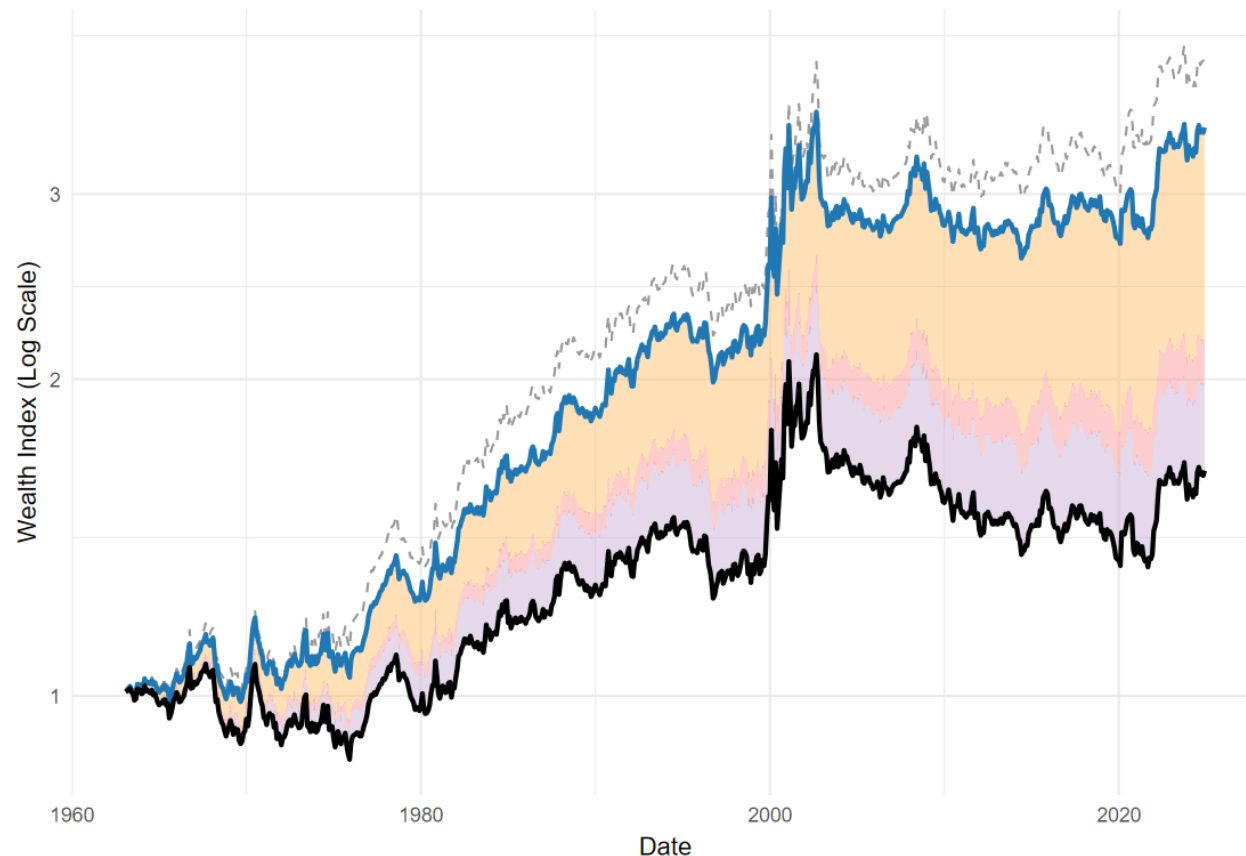


- After all frictions, the **Sharpe ratio decreases to 0.2**.
- Post-2000 the strategy performance flats out.

Limited number of positions – (Top 5% long-short)

Strategy Performance: Decomposition of Frictions (Top 10% Portfolio)

Cumulative wealth lost to Transaction Costs, Borrow Fees, and Dividend Taxes



Strategy Metrics	Theoretical	Gross (Lagged)	Net (Final)
Ann. Return	0.0228	0.0203	0.0080
Ann. Volatility	0.0835	0.0816	0.0816
Sharpe (Rf=0%)	0.2735	0.2488	0.0980



- We try to **limit the number of traded stocks** by taking the top 5% long/short positions only.



- Absolute performance (as well as risk-adjusted measures) **halves** and the performance is hit significantly.

Conclusions

- **Factor Momentum:** Is a good strategy but not be the main strategy of the portfolio
- **Well defined Limits and Universe : Weights, Market Cap sizing, Maybe would work well in the european market**
- **Long Only performs best:** Short positions face a "deadweight loss." Manufactured dividends paid to lenders are non-deductible for private investors due to the Austrian income-related expense deduction ban (§20 Abs 2 EStG), creating a significant "negative alpha" hurdle.
- **Using all of the factors published reduces alpha:** Dividends on the long-leg side must be tax-adjusted to account for the tax differences of long/short positions.
- Turnover

We propose to implement a parameter to asymmetrically tax dividends (27.5% rate) received on the long leg, while paying 100% of the manufactured dividends on the short leg.

Lessons Learnt

Single Stock Implementation

- Factor-level backtests overstated implementability; moving to single-name replication exposed concentration and idiosyncratic risk that the factor series masked.

Processing Power Required

- Security-level replication is computationally intensive; rolling estimations, rebalances, and cost modelling require scalable workflows and efficient storage of large Parquet datasets. The WU cluster and JKP repository were essential for running the depth-of-analysis and robustness checks we targeted.

Frictions Implementation

- Transaction Costs: Transaction costs and short financing are material at scale. Once implemented, a significant share of gross alpha was absorbed by trading costs and shorting expenses.
- Lagging the signal by one trading day reduced implementation bias and more accurately reflected real execution/availability of inputs, though it weakened raw back-test performance.

Empirical Implementation Reality Check

- This project bridged factor momentum research and real-world implementation at scale, highlighting how portfolio construction choices and frictions (lag, turnover, borrowing, and transaction costs) can turn a clean academic effect into a challenging strategy to implement.

R Shiny App

- More interactively what we did:
- https://huggingface.co/spaces/farkastallos/llab_Factor_Momentum

Credentials:

- User: llab_member
- Password: llab_member_12_1

Quant Lab: Dynamic Momentum vs Benchmarks

1. Configuration

Select Analysis Period:

2005-01-01 to 2025-12-03

2. Momentum Construction

Formation Lookback (Months) [J]:

1 12 24

Skip/Lag Period (Months):

0 1 3

Holding Period (Months) [K]:

1 3 5 7 9 11 12

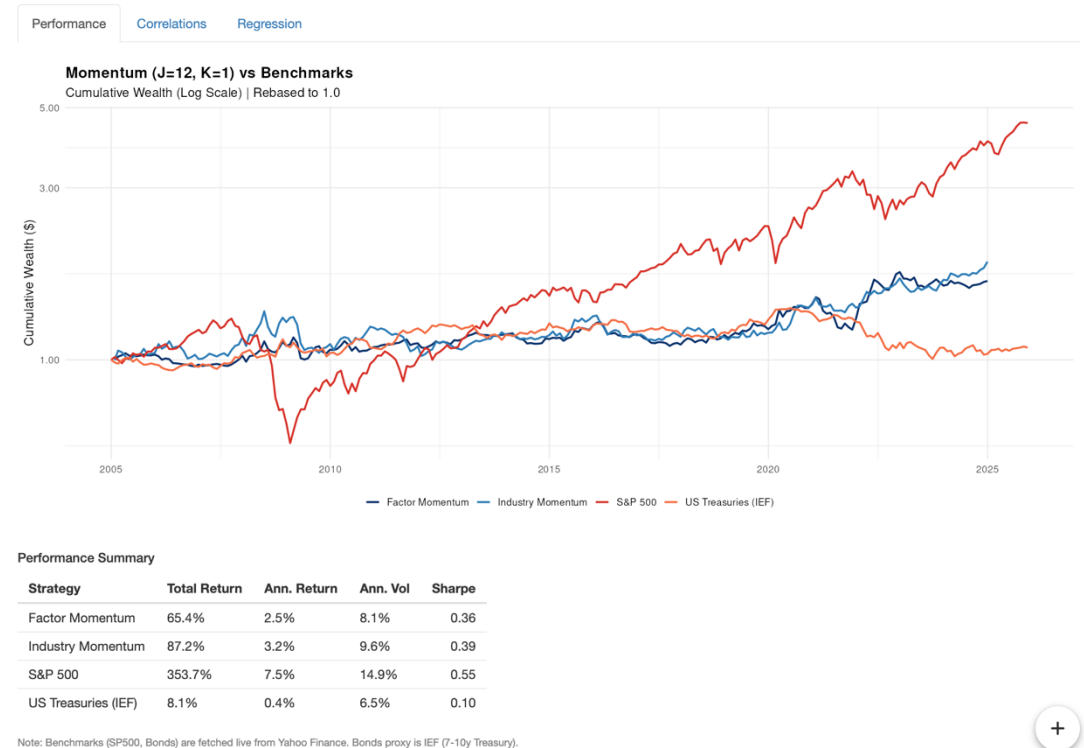
3. Regression Settings

Target Factor:

Book_to_Market_HML

FF Model Factors:

☒ mkt_excess
 ☒ smb
 ☒ hml
 ☐ rmw
 ☐ cma
 ☐ mom



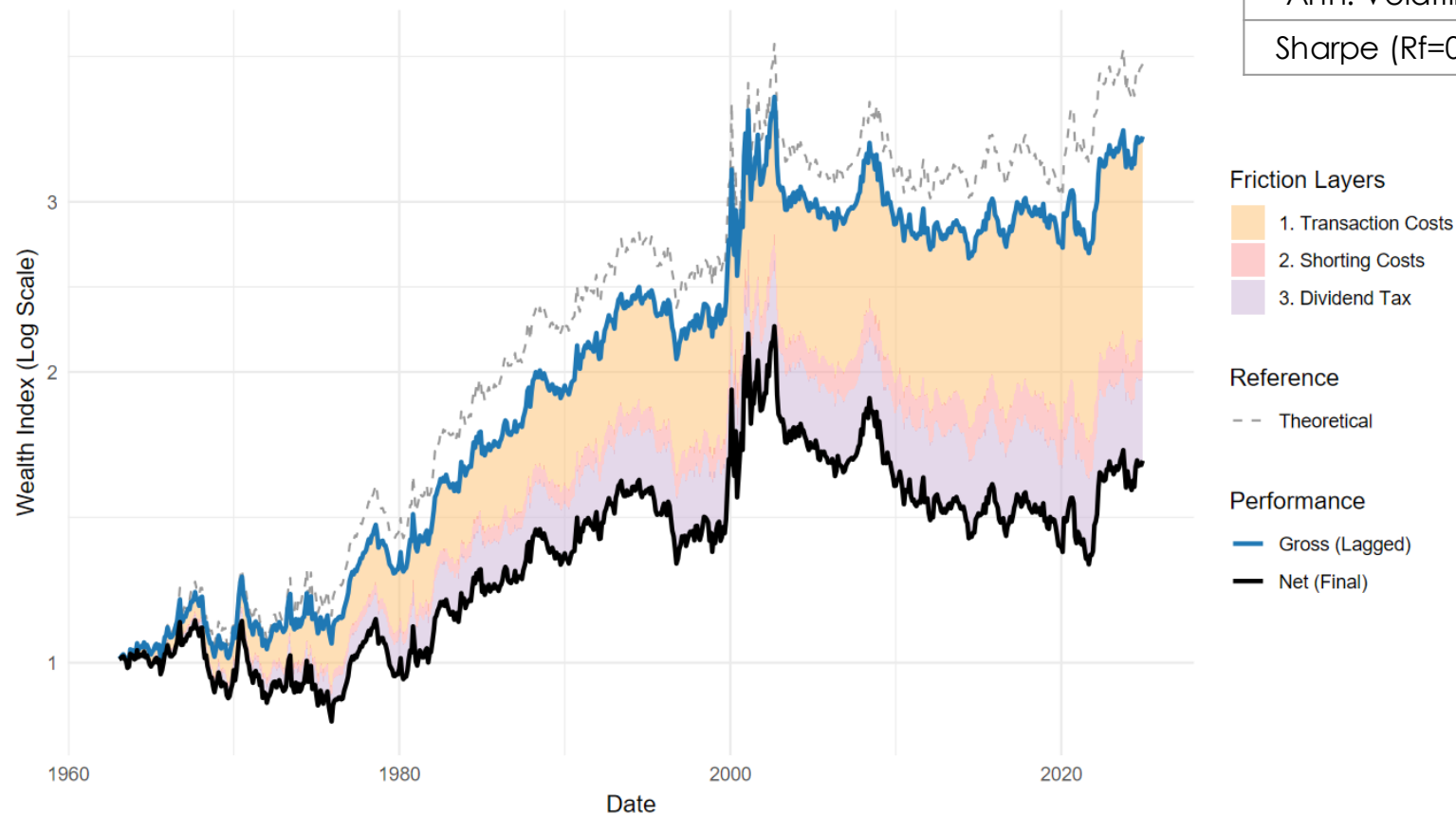
Thank you!



Limited number of positions – (Top 2.5% long-short)

Strategy Performance: Decomposition of Frictions (Top 5% Portfolio)

Cumulative wealth lost to Transaction Costs, Borrow Fees, and Dividend Taxes



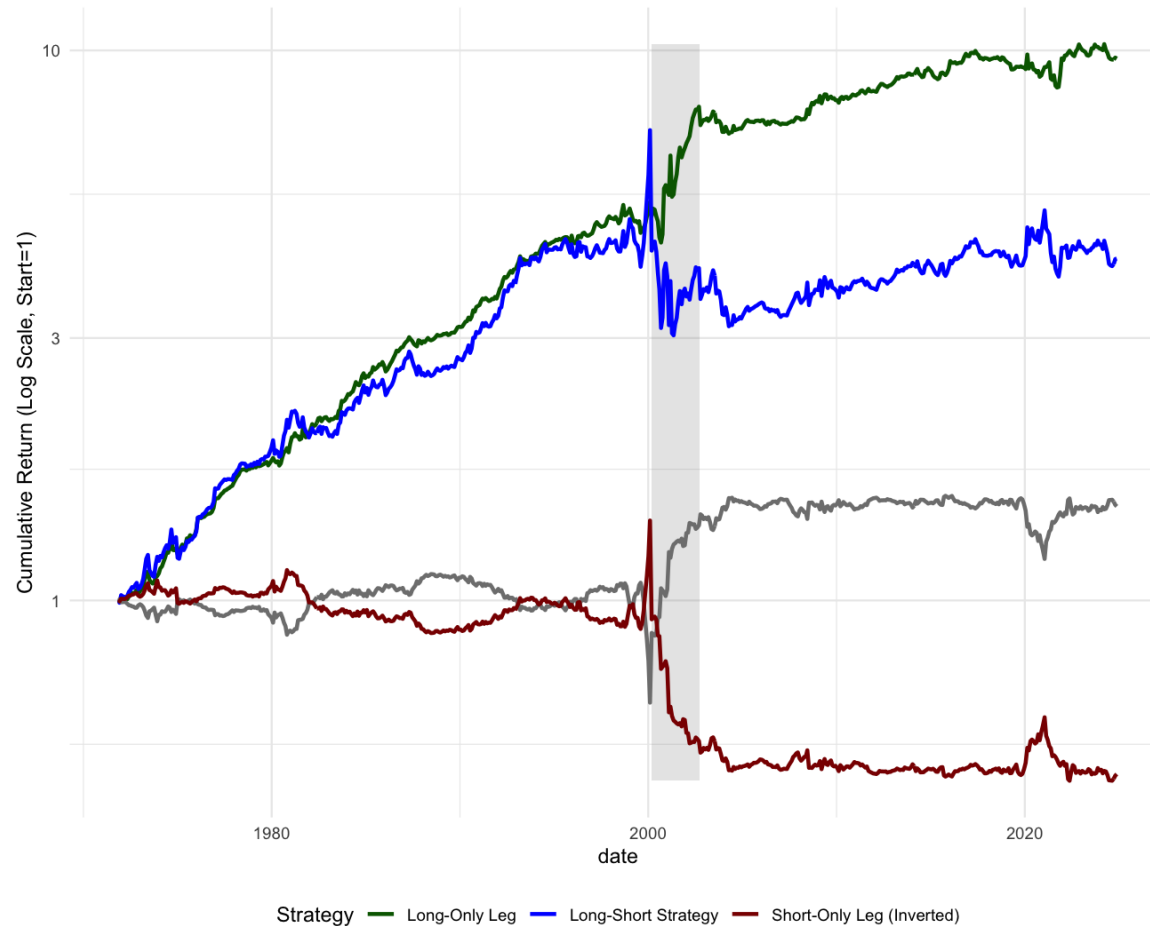
Strategy Metrics	Theoretical	Gross (Lagged)	Net (Final)
Ann. Return	0.0233	0.0205	0.0078
Ann. Volatility	0.0927	0.0907	0.0907
Sharpe (Rf=0%)	0.2515	0.2257	0.0861

Factor Regression on FF3 & FF5+Momentum

- We regressed all the factors first on FF3, then on FF5 + Momentum
- We used this step to identify factors to exclude (e.g., insignificant alpha, beta of approx. 1)
- This will be shown better in the next slides.

Lagged Signals

Factor Momentum (21d Lookback, Monthly Rebal) Signal Lagged 2 days
Mimicking Monthly Data using Daily inputs



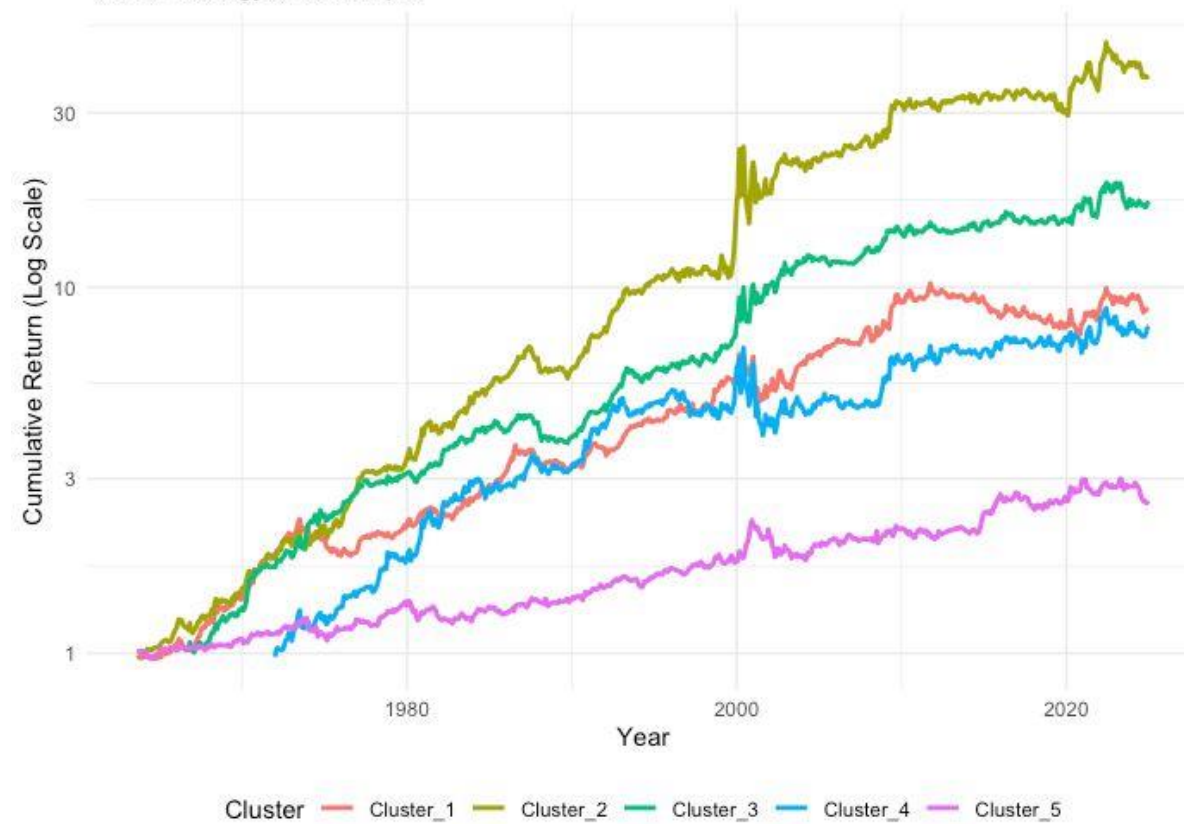
- We implement a 21-day lookback, 21-day holding factor-momentum strategy, but lag the signal by 2 trading days.
- At each monthly rebalance, the portfolio is formed using information only up to T-2.

Clustering Performance

Cluster Theme	Count	Factors
Size / Microstructure / Trading	6	Size_SMB, Long_Term_Reversals_LTREV, Quality_Minus_Junk_QMJ, Amihud_Illiquidity, Nominal_Price, High_Volume_Premium
Valuation & Investment Intensity	11	Book_to_Market_HML, Asset_Growth_CMA, Cash_Flow_to_Price, Earnings_to_Price, Enterprise_Multiple, Sales_to_Price, Growth_in_Inventory, Sales_Growth, CAPX_Growth_Rate, Leverage_Factor, Altman_Z_Score
Quality & Earnings Quality	9	Operating_Profitability_RMW, Gross_Profitability, Return_on_Equity, Return_on_Assets, Change_in_Asset_Turnover, Accruals_Factor, Net_Working_Capital_Changes, Debt_Issuance_Factor, Ohlson_O_Score
Low-Risk & Financing / Lifecycle	7	Residual_Variance_RVAR, Low_Beta_BAB, Firm_Age, Profit_Margin, One_Year_Share_Issuance, Total_External_Financing, Piotroski_F_Score
Asset Expansion (NOA)	3	Net_Operating_Assets, Growth_in_Sales_Inventory, Abnormal_Investment

Clustering Performance

Comparison of Factor Momentum: Long-Short Performance by Cluster
1M/1M Strategy, Median Split



We see Valuation & Investment Intensity being the main driver

Performance during Crises

	Factor_Mom_Return	Market_Return
COVID (2020)	0.0801	-0.204
GFC (2007-2009)	1.06	-0.461
Correlation in bad market months ($\text{Mkt-Rf} < -2\%$)		-0.016

The Factor Momentum Strategy acts as a powerful diversification tool, especially during market crises

Correlation to 60/40 Portfolio

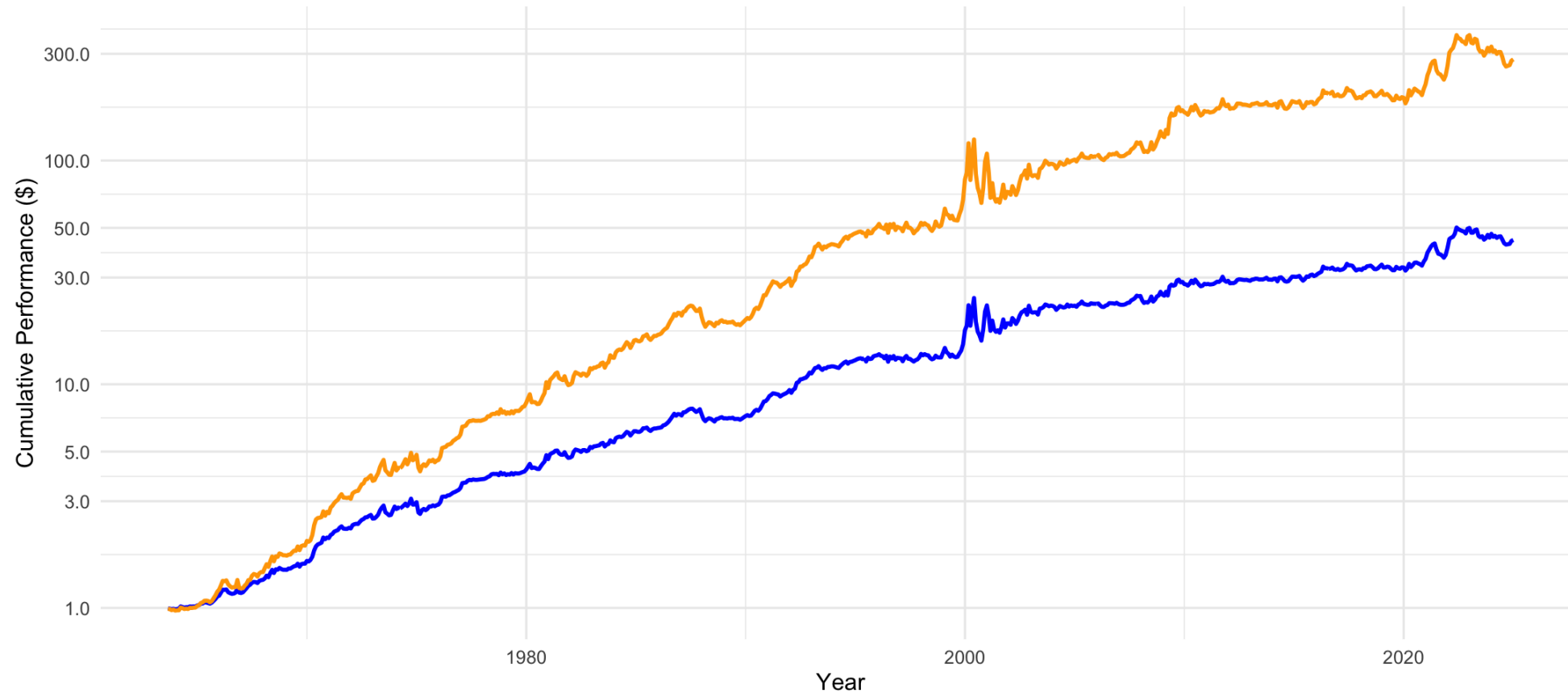
- **Corr = -0.034**

The strategy provides diversification benefits and reduces dependence on traditional risk premia.

Change of Momentum Definition (Long top 25%, Short bottom 25%)

Factor Momentum Comparison (Raw Returns)
50/50 (Median) vs. 25/25 (Quartile) Strategy | Log Scale

Strategy — Factor Momentum — Factor Momentum (Quartile)



Factor selection

Selection Criteria:

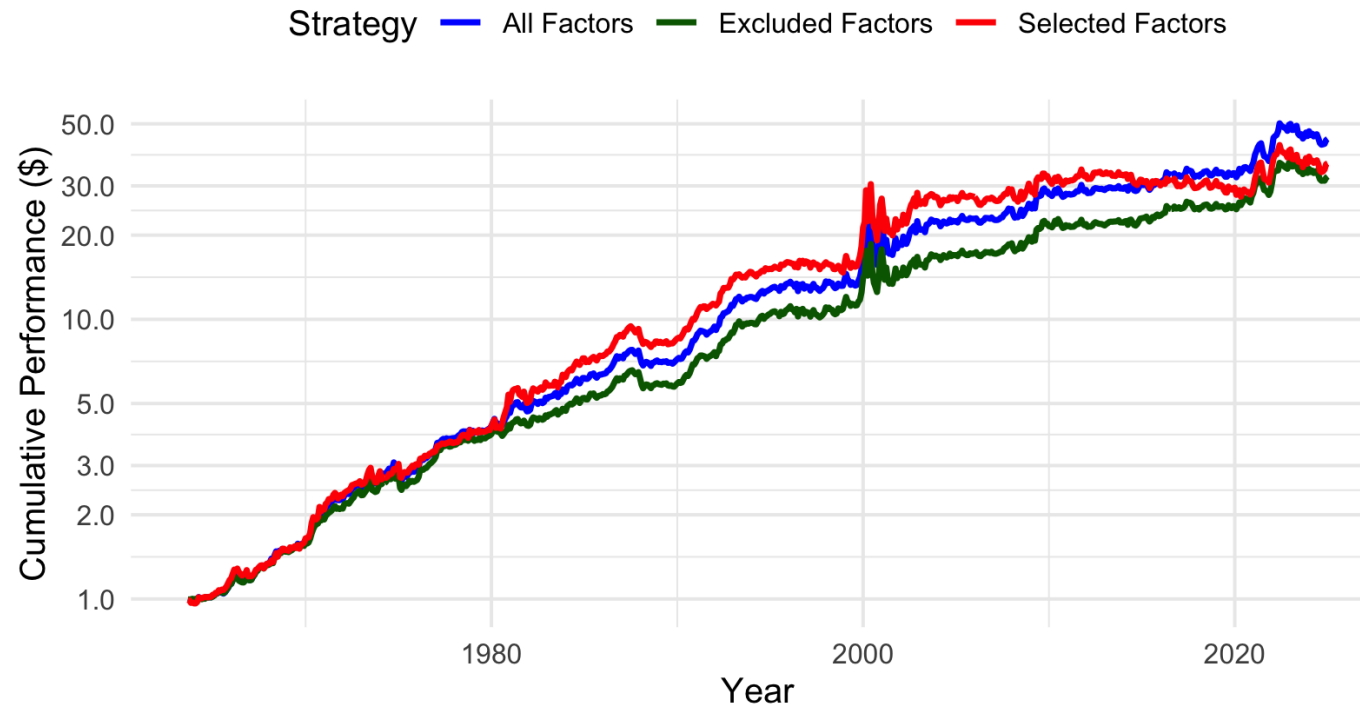
- **Statistical Significance:** keep factors with a significant alpha.
- **Thematic Uniqueness:** The factor must provide unique information and not be highly correlated with (or thematically redundant with) other selected factors. If multiple factors represented the same theme (e.g., "Value" or "Profitability"), only one was kept.
- We also included the standard Fama French factors.

Selected Factors:

1. Size_SMB
2. Book_to_Market_HML
3. Operating_Profitability_RMW
4. Asset_Growth_CMA
5. Long_Term_Reversals_LTREV
6. Low_Beta_BAB
7. Firm_Age
8. Enterprise_Multiple
9. Accruals_Factor
10. Total_External_Financing
11. Piotroski_F_Score

Selected factors vs. all

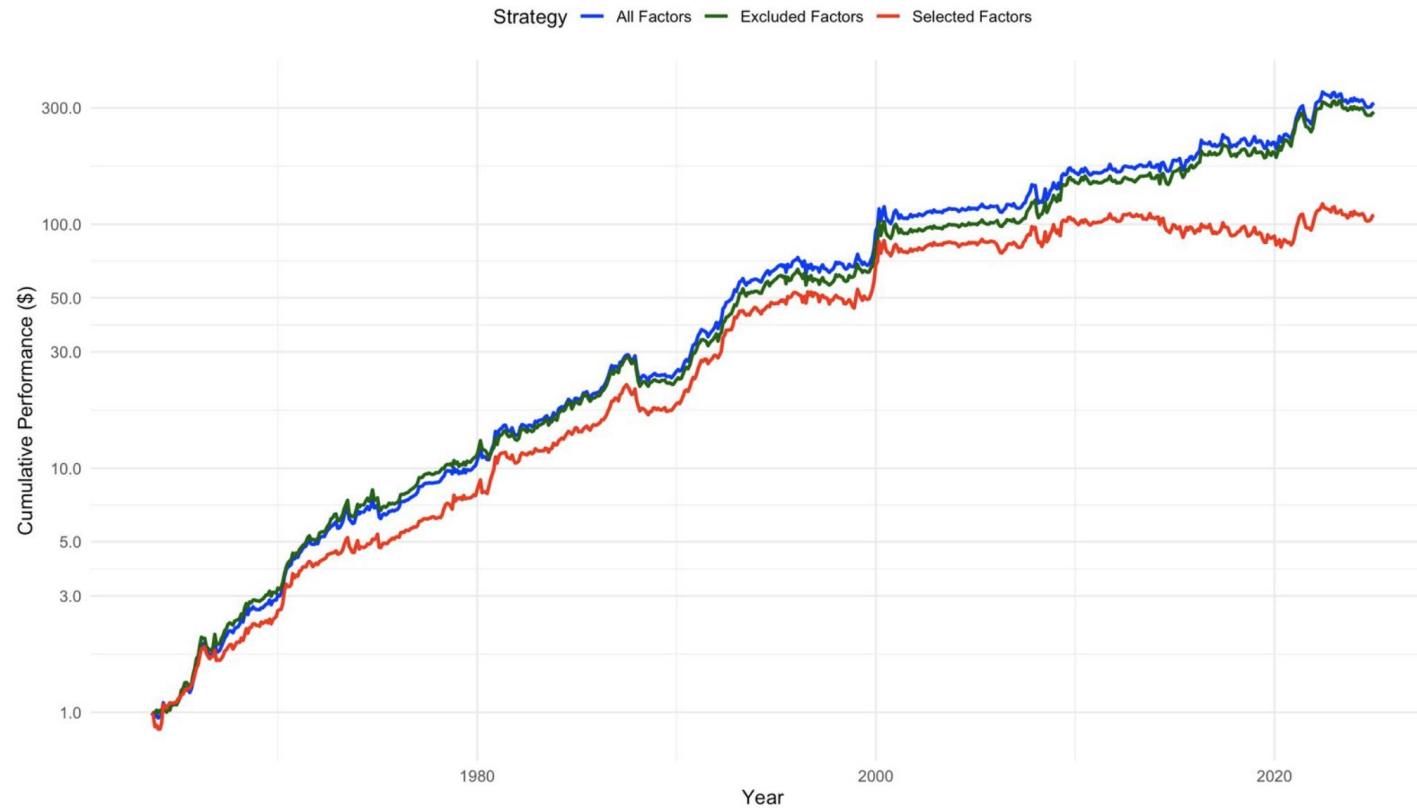
Cumulative Performance: All vs. Selected vs. Excluded Factors (Unscale
Value of \$1 invested (Log Scale))



Warning: once we use a volatility scaled strategy, the selected factors underperform the other two.

Selected factors vs. all

Cumulative Performance: All vs. Selected vs. Excluded Factors (Scaled to 10% Ann. Volatility)
Value of \$1 invested (Log Scale)



Warning: once we use a volatility scaled strategy, the selected factors underperform the other two.

Selected factors vs. industry

Cumulative Performance of Selected Factor Momentum vs. Industry Momentum (Unscaled)
Value of \$1 invested (Log Scale)

